

The functional analytic aspects of operator folding and finite element methods

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In [4] Colbrook introduces a very appealing algorithm for computing spectra, and more interestingly pseudospectra, of differential operators. This note makes two claims about its relation to the finite element method.

(a) *The algorithm as stated in [4, §3.4.1] cannot be followed by a finite element method.* It is phrased in terms of A , A^* , $D(A)$, $D(A^*)$ and orthogonal projections $\mathcal{P}_n$, and a finite element code has none of those. It has a bounded sesquilinear form on $V \times V$. Section 2 lists six places where the passage from one to the other breaks down.

(b) *What is actually computed for Maxwell in [4, §3.4.4] is not that algorithm.* It is, we believe, the functional analytic setting we set out in section 3, namely a conforming Rayleigh–Ritz discretisation of the pencil $(**) u = \lambda Mu$ on $V \times V$. Section 3.2 gives our evidence.

If (b) is right the two are reconciled, and what is missing from [4] is the bridge between them, which is section 3.

1 The setting, as we read it

Folding starts from a closed, densely defined A on a Hilbert space $\mathcal{H}$. Since we want its spectrum, A must be an *endomorphism*, $A: D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$, the classical instance being $\Delta: H^2(\Omega) \cap H_0^1(\Omega) \rightarrow L^2(\Omega)$. Under suitable conditions on $D(A)$ and $D(A^*)$ one considers [4, §3.4.1]

$$\gamma(z, A) := \min\{\sigma_{\inf}(A - zI), \sigma_{\inf}((A - zI)^*)\} = \|(A - zI)^{-1}\|^{-1}, \quad (1)$$

where the injection moduli come from the quadratic forms of the resolvent and its adjoint,

$$\sigma_{\inf}(A - zI)^2 = \inf_{f \in D(A) \setminus \{0\}} \frac{\|(A - zI)f\|_{\mathcal{H}}^2}{\|f\|_{\mathcal{H}}^2}, \quad \sigma_{\inf}((A - zI)^*)^2 = \inf_{f \in D(A^*) \setminus \{0\}} \frac{\|(A - zI)^*f\|_{\mathcal{H}}^2}{\|f\|_{\mathcal{H}}^2}. \quad (2)$$

So far everything is formal and we are happy. Our difficulty begins with the comment below [4, Eq. (3.26)], that this reduces the computation of pseudospectra to the eigenvalues of the self-adjoint operator

$$(A - zI)^*(A - zI). \quad (3)$$

2 Why we cannot follow the algorithm

(i) **The conditions on $D(A)$ and $D(A^*)$ are left unstated.** The sentence introducing [4, Eq. (3.25)] reads “*under suitable conditions on $D(A)$ and $D(A^*)$, the functions $\Phi_n(z, A)$ converge locally uniformly to $\gamma(z, A)$ from above and hence satisfy Assumption 3.2.1*”. That hypothesis carries everything, since convergence from above is what Assumption 3.2.1 asks for and what turns [4, Thm 3.2.3] into $\Gamma_n^\epsilon(A) \subseteq \text{Sp}_\epsilon(A)$, the error control. We cannot tell what it is. Reading back from [4, Alg. 3.4], which asks for vectors whose span is a core of A and of A^* , we take it to be

$$\text{ran}(\mathcal{P}_n^*) \subseteq D(A) \cap D(A^*) \text{ for each } n, \quad \bigcup_n \text{ran}(\mathcal{P}_n^*) \text{ a core of } A \text{ and of } A^*, \quad (4)$$

the first half being needed for Φ_n to be defined at all. On that reading one and the same finite-dimensional space must lie in *both* domains, and those carry different boundary conditions in general.

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If instead the conditions are meant to make (3) well posed, they are different ones. Equation (3) is the operator T^*T of [6, §V.3.7], and by [6, Thm V.3.24, p. 275] it needs the strictly smaller $D(T^*T) = \{u \in D(T) : Tu \in D(T^*)\}$ with $T = A - zI$, which a discretisation would have to respect instead.

(ii) Babuška–Osborn does not apply. We would like to apply Babuška–Osborn theory [1, 3] to (3), and cannot, since it is unbounded with non-compact inverse. We take this to be deliberate, as [4, p. 129] says “it is not necessary to assume that the operator has a compact resolvent”. What we cannot see is what plays its role. Rayleigh–Ritz on a conforming subspace bounds γ from above for free, which we take to be the source of the error control, but it gives no rate and no local indicator, and [4, Prop. 3.3.12] gives both only for *normal* A with an isolated eigenvalue and a computable gap. Non-normality is the case pseudospectra exist for.

(iii) A primal-dual mismatch. Read (3) as what it looks like, a product of $A - zI$ with its adjoint. The Banach adjoint is $(A - zI)^* : \mathcal{H}^* \rightarrow \mathcal{H}^*$, so we would be composing $\mathcal{H} \xrightarrow{A-zI} \mathcal{H}$ with $\mathcal{H}^* \xrightarrow{(A-zI)^*} \mathcal{H}^*$, and the two halves do not meet. The standard repair, when $\mathcal{H} = L^2$, is to identify $\mathcal{H}$ with its dual silently. That does keep the composition an endomorphism. We are not fond of it.

(iv) What the identification hides. It makes the Riesz map coincide with the identity. Writing $M : \mathcal{H} \rightarrow \mathcal{H}^*$ for the map it really is, the operator ought to be

$$M^{-1} (A - zI)^* M (A - zI), \quad (*)$$

that is $\mathcal{H} \xrightarrow{A-zI} \mathcal{H} \xrightarrow{M} \mathcal{H}^* \xrightarrow{(A-zI)^*} \mathcal{H}^* \xrightarrow{M^{-1}} \mathcal{H}$. Undoing the identification therefore only exhibits the Hilbert adjoint as $M^{-1}(\cdot)^* M$ applied to the Banach adjoint, and (*) is (3) again, since $(M^{-1}(A - zI)^* M (A - zI)u, v)_{\mathcal{H}} = \langle M(A - zI)u, (A - zI)v \rangle = ((A - zI)u, (A - zI)v)_{\mathcal{H}}$. So we are no better off. Finite element operators go into $\mathcal{H}^*$, not $\mathcal{H}$, and we do not see how to read a variational formulation off (*).

(v) The trial space must be a core in the graph norm. Points (i) to (iv) are local. The structural difficulty is that folding evaluates $\|(A - zI)u\|_{\mathcal{H}}$, so the trial space must sit inside the *graph* space $D(A)$ and exhaust a core of it. For Δ on a bounded Ω four things then go wrong at once. Continuous piecewise linears are not in $H^2(\Omega)$, so $\|(A - zI)u_h\|_{L^2}$ is not inaccurate but undefined, and C^1 elements would be needed. On a non-convex polygon even those fail, since $D(\Delta) = \{u \in H_0^1 : \Delta u \in L^2\}$ contains corner singularities and has $H^2 \cap H_0^1$ as a proper *closed* subspace in the graph norm, so no H^2 -conforming family is a core. Nor is $C_c^\infty(\Omega)$ a core, its graph closure being $H_0^2(\Omega)$, so the reflex that smooth functions are dense is unavailable. And $D(A^*)$ carries different boundary conditions, so the second quotient in (2) needs a second space. None of this arises in [4, §§3.4.2–3.4.3], set on $L^2(\mathbb{R}^d)$ where $C_c^\infty(\mathbb{R}^d)$ is *assumed* to be a core and the basis is Hermite functions, which are smooth, with no boundary. Nor in [4, §3.4.4], whose operator is the *first-order* Maxwell system, whose graph space $H_0(\text{curl}) \times H(\text{curl})$ standard elements reach. Between them the two settings never cover a second-order operator on a bounded domain with boundary conditions. The same conclusion is reached from the discrete side in [7, Rmk 3.7], which warns that the L^2 Riesz map “will not be consistent with the function spaces of the continuous problem”.

(vi) The estimator we want needs more than either. We intend to drive refinement by a dual weighted residual estimator [2, 5]. DWR needs a form bounded on $V \times V$, a conforming V_h for Galerkin orthogonality, a well posed dual problem, and, for an eigenvalue problem, an isolated eigenvalue to linearise about. The third is free, the folded operator being self-adjoint so that the dual problem is the primal one. The first two are (v) again. The fourth is the hard one. With *essential* spectrum at the bottom there is no eigenfunction, hence no dual problem and nothing to linearise about. The discrete minimiser still exists and the upper bound of (ii) still holds, so the certificate survives, but the estimator has no anchor. The tool needs precisely the isolation that [4, p. 129] declines to assume. Whether the bottom is isolated also depends on where z sits, so the inner and outer layers of adaptivity are not independent.

3 The setting we propose

Let $V \subset \subset \mathcal{H}$ with $\mathcal{H} = L^2$, taking V of the type $H_0(\text{curl}) \cap H(\text{div } 0)$ so the embedding is compact.¹ From here A is the *weak* operator, related to the endomorphism of section 1 by $A_s = M^{-1}A$, and only now does the shift $A - zM$ make sense. Take

$$A: V \times V \rightarrow V^* \times V^*, \quad \langle A \begin{pmatrix} E \\ H \end{pmatrix}, \begin{pmatrix} F \\ G \end{pmatrix} \rangle = (-i \bar{\nabla} \times E, F)_{L^2} + (i \hat{\nabla} \times H, G)_{L^2}. \quad (5)$$

Since A is *first order* it goes not into V^* but into $\mathcal{H}^* \subseteq V^*$. So we may bring in the Riesz map $M^{-1}: \mathcal{H}^* \rightarrow \mathcal{H}$ and write the folded operator as

$$(A - zM)^* M^{-1} (A - zM), \quad (**)$$

which composes, $V \times V \xrightarrow{A-zM} \mathcal{H}^* \times \mathcal{H}^* \subseteq V^* \times V^* \xrightarrow{M^{-1}} \mathcal{H} \times \mathcal{H} \cong (\mathcal{H}^* \times \mathcal{H}^*)^* \xrightarrow{(A-zM)^*} V^* \times V^*$, the middle step being the canonical double dual identification, numerically an identity. This is the continuous counterpart of the modified normal equation $A^T T A x = A^T T b$ of [7, Eq. (3.7)], where T discretises a Riesz map inserted for the reason given in section 2(iii), that the plain normal equation acts on inconsistent function spaces. Folding adds that the Riesz map is not free. It must be that of the pivot space $\mathcal{H}$, where σ_{inf} is measured. Unlike (*), this has a variational form attached,

$$a(E, F, H, G) = (\bar{\nabla} \times E, \bar{\nabla} \times F) + (\hat{\nabla} \times H, \hat{\nabla} \times G) - (\bar{z} + z) i (\hat{\nabla} \times H, F) + (\bar{z} + z) i (\bar{\nabla} \times E, G) + \bar{z} z (E, F) + \bar{z} z (H, G). \quad (6)$$

3.1 The Gårding inequality, and the standard finite element device

Section 2 got stuck because [4] needs an endomorphism while a finite element method produces $A: V \rightarrow V^*$, an endomorphism of nothing, with no spectrum of its own. Above we repaired that by *composing* with a Riesz map, recovering the unbounded $A_s = M^{-1}A$ on $\mathcal{H}$. A finite element method usually does the opposite and *inverts*, and what comes out is compact rather than unbounded. Two steps, the first free.

A Gårding inequality is *automatic*, with no hypothesis beyond V carrying the graph norm $\|u\|_V^2 = \|u\|_{\mathcal{H}}^2 + \|A_s u\|_{\mathcal{H}}^2$. Applying $(s - t)^2 \geq \frac{1}{2} s^2 - t^2$ with $s = \|A_s u\|_{\mathcal{H}}$ and $t = |z| \|u\|_{\mathcal{H}}$ gives, for $u \in V$,

$$a(u, u) = \|(A_s - zI)u\|_{\mathcal{H}}^2 \geq \frac{1}{2} \|A_s u\|_{\mathcal{H}}^2 - |z|^2 \|u\|_{\mathcal{H}}^2 = \frac{1}{2} \|u\|_V^2 - \left(\frac{1}{2} + |z|^2\right) \|u\|_{\mathcal{H}}^2, \quad (7)$$

so $a + C(\cdot, \cdot)_{\mathcal{H}}$ is coercive on V for $C = \frac{1}{2} + |z|^2$, the constant $\frac{1}{2}$ being unimportant beyond its independence of z .

Lax–Milgram then makes $(**) + CM: V \rightarrow V^*$ an isomorphism, so the solution operator

$$T: \mathcal{H} \rightarrow V, \quad a(Tf, v) + C(Tf, v)_{\mathcal{H}} = (f, v)_{\mathcal{H}} \quad \text{for all } v \in V, \quad (8)$$

is well defined, bounded, self-adjoint and positive. Composed with $V \subset \subset \mathcal{H}$ it is a *compact endomorphism* of $\mathcal{H}$, and this, not (**), is the object whose spectrum a finite element method studies. Its nonzero eigenvalues are reciprocal to those of the pencil, $\lambda = \mu^{-1} - C$, so Riesz–Schauder gives a pencil spectrum $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \rightarrow +\infty$ of finite multiplicities, equivalently $(**) - \lambda M$ Fredholm of index zero for every λ . There is no essential spectrum, the bottom λ_1 is an isolated eigenvalue, equal to $\sigma_{\text{inf}}(A_s - zI)^2$ when V is the whole graph space, and Babuška–Osborn theory [1, 3] applies to T in the usual way, bringing Kato–Temple and hence DWR with it.

The device does not solve the difficulty of section 1, it inverts it. An endomorphism is produced by *solving* rather than by *applying*, trading the unbounded A_s for the compact T with reciprocal spectrum. The Gårding inequality is what makes T exist, the compact embedding is what makes it compact.

¹Two caveats. We are sloppy about the components: E and H carry different boundary conditions, so the pair lives in $H_0(\text{curl}) \times H(\text{curl})$ rather than $V \times V$. And the solenoidal constraint buys the compact embedding at the cost of conformity, since standard elements are not divergence-free, as flagged in [4, footnote 2, p. 130].

We know what this costs. A compact embedding $V \subset\subset \mathcal{H}$ of the graph space is the same thing as compactness of the resolvent of A_s , exactly what [4, p. 129] declines to assume. For Maxwell it forces the solenoidal V of the footnote, which standard finite elements do not provide. That trade seems to us the crux of the whole question.

3.2 This appears to be what is already done for Maxwell

Read [4, §3.4.4] back through the above. The quantity minimised in step 2 on [4, p. 132] is

$$\min_{(E,H) \in V_N \setminus \{0\}} \frac{\int_U |\hat{\nabla} \times H - zE|^2 + |-i\bar{\nabla} \times E - zH|^2 dx}{\int_U |E|^2 + |H|^2 dx} = \min_{u_h \in V_N \setminus \{0\}} \frac{\|(A_s - zI)u_h\|_{L^2}^2}{\|u_h\|_{L^2}^2}, \quad (9)$$

over a conforming $V_N \subset V$ of *plain Lagrange elements*. Choosing a basis of V_N , the numerator is the stiffness matrix of (6), that is of (**), and the denominator is the mass matrix. So (9) is precisely the Rayleigh–Ritz discretisation of the pencil (**) $u = \lambda Mu$.

What is never built is as telling as what is. The operator (3) is not formed. No adjoint is applied to anything. No domain of Kato type is discretised. No projection $\mathcal{P}_n$ appears. Only a bilinear form and a mass matrix are assembled, which is all a finite element code can offer. Since the Maxwell operator is self-adjoint the second quotient in (2) coincides with the first and is skipped, which is why the two-space difficulty of section 2(i) does not surface there either.

This is why we suspect the algorithm as stated in [4, §3.4.1] is not what a finite element method can follow, and that [4, §3.4.4] follows (**) instead. We are not suggesting that section is wrong. We are suggesting it does something the statement of the algorithm does not say, and that the missing step is section 3.

4 Open questions

1. **The main one.** Is the computation in [4, §3.4.4] already the reading (**), rather than the algorithm of [4, §3.4.1]? If so, is the passage between the two the one given in section 3, and can the algorithm be restated in those terms?
2. Which reading of section 2(i) is meant, (4), so that only the forms (2) on $D(A)$ and $D(A^*)$ are ever needed, or Kato's $D(T^*T)$, so that (3) itself must be realised?
3. Is the mismatch of section 2(iii) real, and is the Riesz map in (**) the right repair? Is it forced to be the Riesz map of the pivot space $\mathcal{H}$ rather than that of V ?
4. Section 3.1 removes the essential spectrum that blocks DWR in section 2(vi), at the price of the compact embedding, which is compactness of the resolvent and is what [4] deliberately avoids. Is that trade the right one when the goal is a finite element method with a computable local error indicator? If not, what drives refinement when the bottom of the folded spectrum is essential?

References

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